Venkata Lakshmi Parimala Pasupuleti

Software Engineer · Agentic AI · Human-in-the-Loop · RAG Pipelines · MCP · Python · TypeScript · AWS · Enterprise Automation

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github.com/venkatalakshmiparimala | Virginia, USA (Remote)

PROFESSIONAL SUMMARY

Software engineer with 9+ years of production experience who has designed and shipped agentic AI systems with human-in-the-loop controls, RAG pipelines, and enterprise automation workflows end-to-end. Leasel is the core proof point for this role: I designed the full HITL architecture - a multi-step agentic workflow (8 LangGraph agents with conditional routing) where an AI safety gate escalates low-confidence decisions to human reviewers before any regulated action is taken, exactly the pattern Smartsheet's Automation team uses with AI-driven decisions and Action Center. AgentProbe completes the picture: a production LLM evaluation platform with vector database RAG (chunking strategies, embedding optimization, hallucination rate tracking), CI-integrated quality gate, and Model Context Protocol (MCP) integration for unified AI-to-system data retrieval. At Entrata, I owned enterprise-grade production systems (AWS, RabbitMQ event-driven architecture, Docker microservices, FastAPI) for 9+ years - 99.5% uptime, zero security incidents, full PHI compliance - translating complex business workflows into automated, auditable software. I use Python and TypeScript as my primary languages and deliver end-to-end: design, build, CI/CD, production support, and root-cause analysis. M.S. Data Analytics Engineering, George Mason University (GPA 3.87, May 2026).

TECHNICAL SKILLS

Agentic AI & Automation: Python (9yr), TypeScript, JavaScript, LangGraph, LangChain, Agentic Workflows, Multi-Agent Orchestration, Human-in-the-Loop (HITL), Guardrails, Prompt Engineering, MCP (Model Context Protocol), Anthropic API, OpenAI API, Ollama, Claude Code, Generative AI

RAG & Data: RAG Pipelines, Vector Databases, ChromaDB, Embeddings, Chunking Strategies, Semantic Search, LLM Evaluation, LLM-as-Judge, Hallucination Detection, OCR, Document Intelligence, NLP, BERT, spaCy, PyTorch, scikit-learn, PostgreSQL, MySQL, Oracle, Redis, SQL

Infrastructure & Delivery: AWS (S3, EC2, RDS, Glue), Docker, FastAPI, REST APIs, RabbitMQ (event-driven, async workflows), GitHub Actions CI/CD, Production Observability, Root-Cause Analysis, Performance Engineering, PHI/HIPAA Compliance, Agile/Scrum, React.js, Linux, Git

PROJECTS

Leasel: End-to-End HITL Agentic Workflow - Safety Gate Design

HITL · Agentic Workflow · RAG · Production AI

Python · LangGraph (8-agent) · FastAPI · TF-IDF RAG · OCR · SQLite · Claude Code | github.com/venkatalakshmiparimala/Lease_complaince_rag

Designed and built the complete human-in-the-loop architecture for a production agentic AI platform - the same pattern Smartsheet's Automation team implements for high-stakes AI-driven decisions. **HITL safety gate:** 8-agent LangGraph workflow (Triage → Retrieval → Configuration → Historical → Reasoning → Validation → Human Escalation) with conditional routing that escalates low-confidence decisions to human reviewers before any regulated action is taken; deterministic guardrails enforce that AI never auto-resolves compliance or eligibility decisions without human approval. **RAG pipeline:** OCR ingestion, section-aware chunking, TF-IDF retrieval with cosine reranking, 100% citation correctness and 0% unsupported-claim rate. **Enterprise automation:** FastAPI backend, audit logging, production observability, and CI-integrated regression gate. 93.6% precision, 87.9% F1 across 1,240 test cases.

AgentProbe: RAG Optimization Platform with MCP & LLMOps

RAG · MCP · LLMOps · CI Gate · Open Source

Python · FastAPI · LangGraph · ChromaDB · MCP · GitHub Actions · Anthropic API · OpenAI API · Ollama | github.com/venkatalakshmiparimala/AgentProbe

Production LLM evaluation platform with RAG performance optimization and Model Context Protocol (MCP) integration for unified AI-to-system data retrieval - addressing the same RAG and context standardization challenges the Automation team tackles across enterprise systems. Vector database indexing and retrieval optimization (ChromaDB, chunking strategies, embedding selection); 5-metric quality engine tracking hallucination rate, groundedness, correctness, latency, and token efficiency; CI-integrated regression gate blocking quality degradation before deployment; multi-provider LLM routing (Anthropic, OpenAI, Ollama) via clean API abstractions. Zero regressions across 1,240+ test cases. Open-source, enterprise-deployable.

AI on FHIR: Enterprise System Integration - HIPAA-Compliant Production

Enterprise APIs · REST · HIPAA · Full-Stack

Python · Flask · React.js · TypeScript · Docker · BERT · spaCy · LLMs · FHIR REST APIs | github.com/venkatalakshmiparimala/AI_on_FHIR

Full-stack AI system that integrates with enterprise REST APIs (FHIR standard) via NLP and LLM inference - translates natural-language queries into structured API calls, enforces HIPAA-compliant RBAC with authentication and full data isolation, ~60% reduction in manual retrieval time. Demonstrates the enterprise system integration pattern (consuming REST APIs and transforming them into user-facing AI tools) the Automation team builds daily.

EXPERIENCE

Magizhs Tech Solution

May 2025 - Aug 2025

AI/ML Engineer - Agentic AI Production Deployment

- Designed and deployed a production agentic AI system (LangGraph, FastAPI, Python) for an external customer - multi-step workflow with RAG over regulatory documents, validation guardrails, and human review gates; translated ambiguous stakeholder requirements into working automated software delivered in weeks as sole technical owner.

George Mason University

Aug 2024 - May 2026

Graduate Teaching Assistant & Research Associate

- Built ML pipelines (PyTorch, scikit-learn) - $R^2 = 0.977$; designed observability dashboards improving research comprehension 40%; mentored 600+ students in Python, SQL, and system design; translated complex research outputs for non-technical faculty stakeholders.

Entrata - Apartment & Affordable Housing SaaS

Sep 2018 - Aug 2024

Senior Software Engineer · Lehi, UT (Remote)

- **Enterprise-grade event-driven automation at scale** - owned RabbitMQ async processing system for gross-rent updates across thousands of tenant records; designed retry logic, idempotency guarantees, and dead-letter handling that eliminated double-processing failures; built Python/Bash automation pipelines eliminating 80% of manual operational workload; 99.5% uptime across multi-tenant SaaS production.
- **Production observability and root-cause analysis** - built internal debugging tools reducing triage time 40%; owned production incident response across HUD compliance modules; sub-10 bug count maintained 6+ consecutive months through proactive monitoring and disciplined root-cause analysis; 300+ peer code reviews; GitHub Actions CI/CD.
- **Business workflow automation and stakeholder delivery** - 120+ Agile sprints with product, compliance, and operations stakeholders; 20+ full-stack production modules shipped (PHP, Python, PostgreSQL, JavaScript); AWS S3/EC2/RDS/Glue; Docker (50% scalability improvement); 100% PHI compliance; mentoring 5+ engineers; leveraged Claude Code to accelerate delivery.

Entrata India - Pune

Jan 2017 - Sep 2018

Associate Software Engineer → Trainee

- Built backend modules (PHP, Python, PostgreSQL) for production SaaS; data pipeline for 1.6M records; AWS S3 migration resolving 90% of legacy storage constraints.

EDUCATION

M.S. Data Analytics Engineering - George Mason University, Virginia — May 2026 · GPA 3.87

B.Tech Computer Science & Engineering - JNTU Anantapur — 2013-2017